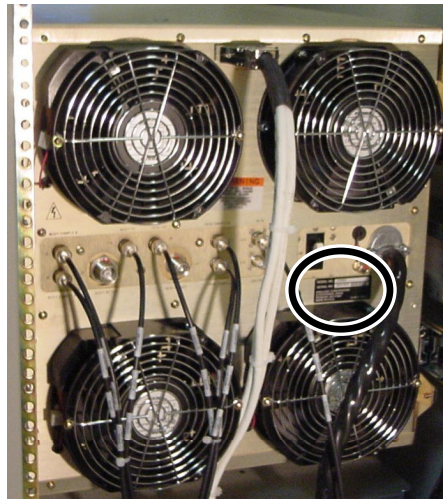


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1- OVERVIEW

This document describes the process for replacing fans on the outside rear of the SRFD2 RF amplifier. There are two different SRFD2 RF amplifier configurations and different fan replacement processes for each. The process for changing fans on older SRFD2 RF amplifiers is much different than that used to change fans on newer units. Older amplifiers can be identified by the serial number printed on the manufacturer rating plate affixed to the rear of the amplifier. See Illustration 1-1.



SRFD2 SERIAL NUMBER RATING PLATE
ILLUSTRATION 1-1

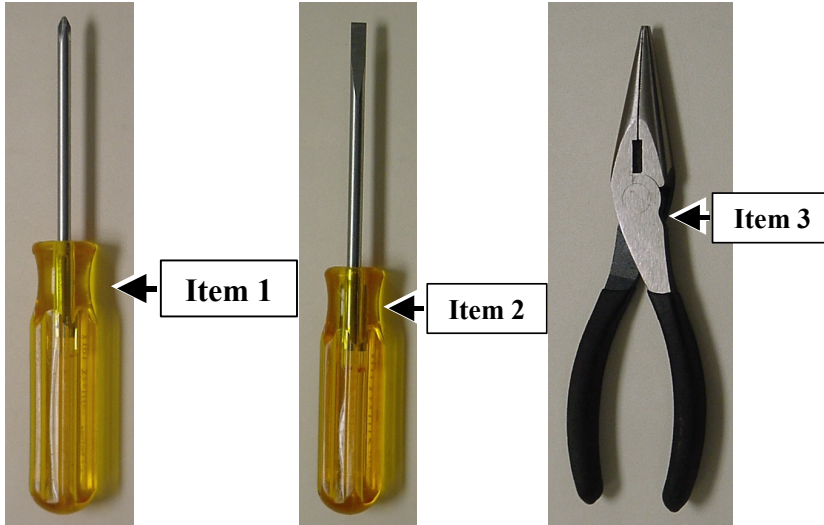
The serial number is printed on the second line of the rating plate and is reported as shown:
YRFW-####

Where **YR** is the last two digits of the year it was built, **FW** is the two digit fiscal week, and **####** is the 4 digit unit number.

Any units where the YR code is **02 or less** and the **####** code is **0001 to 0030** (a letter "P" may be appended after this number) should use the procedure described in **Section 4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY**. Use the procedure described in **Section 4-2 Fan Replacement for Later SRFD2 Amplifiers ONLY** to replace fans in all other units.

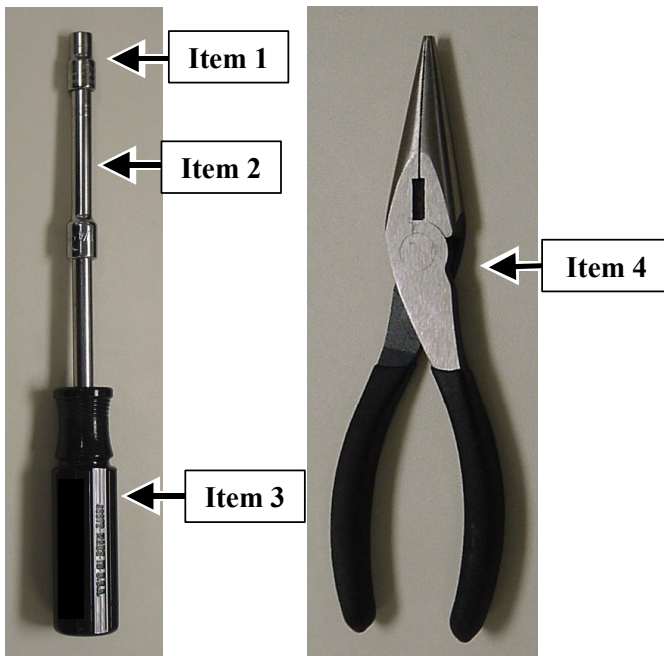
1-1 Tools Required for SRFD2 with Serial Numbers 1 – 30

- #2 Phillips screwdriver (Item 1)
- Slot screwdriver (Item 2)
- Needle nose pliers (Item 3)



1-2 Tools Required for SRFD2 with Serial Numbers 31 or Greater

- 7/32 inch socket with ¼ inch (~0.635cm) drive (Item 1)
- 3 inch long ¼ inch (~0.635cm) drive extension (Item 2)
- ¼ inch (~0.635cm) drive handle (Item 3)
- Needle nose pliers (Item 4)



2- RF CABINET POWER LOCKOUT AND TAGOUT PROCESS FOR ALL AMPLIFIERS

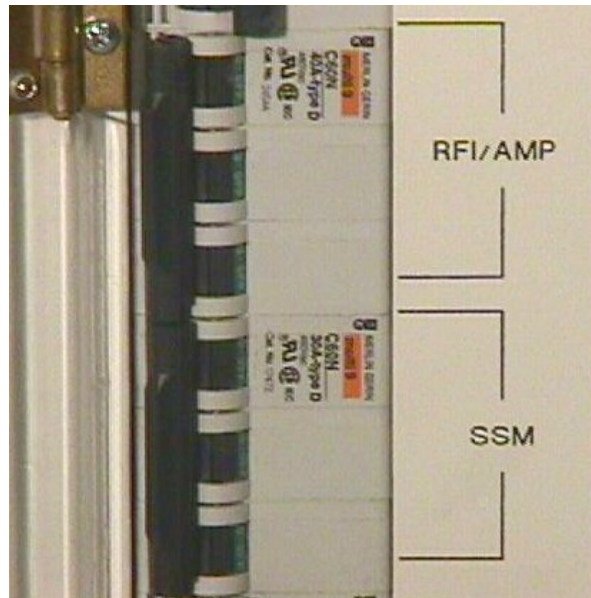


POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE SRFD2 CABINET BEFORE ATTEMPTING TO SERVICE THE UNIT. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

1. Remove the cover from the front of the ACGD cabinet.
2. Open Circuit Breaker Cover on the front of the PDU and switch OFF the **RFI/AMP** and **SSM** breakers. See Illustration 1-1.

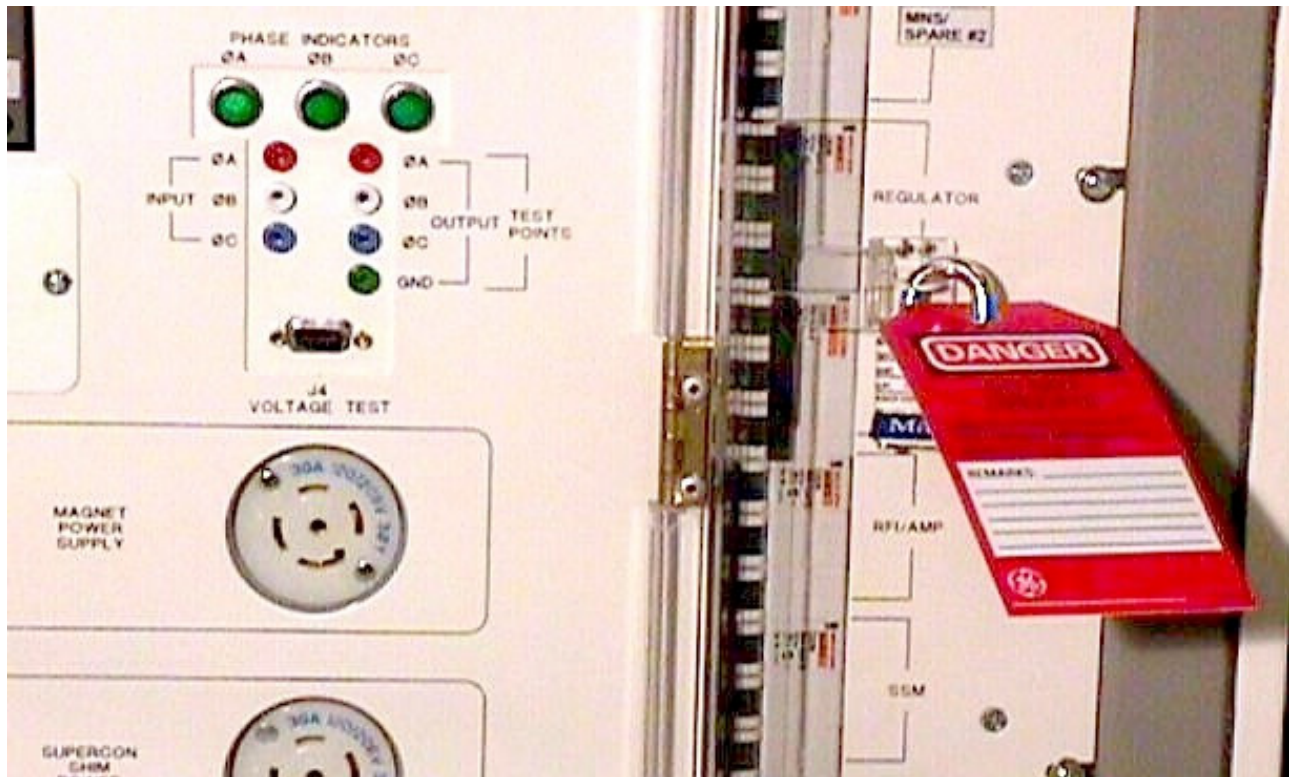
Note

The rest of the system does not need to be shut down for this procedure. Only the RF Cabinet.



BREAKER IDENTIFICATION
ILLUSTRATION 1-1

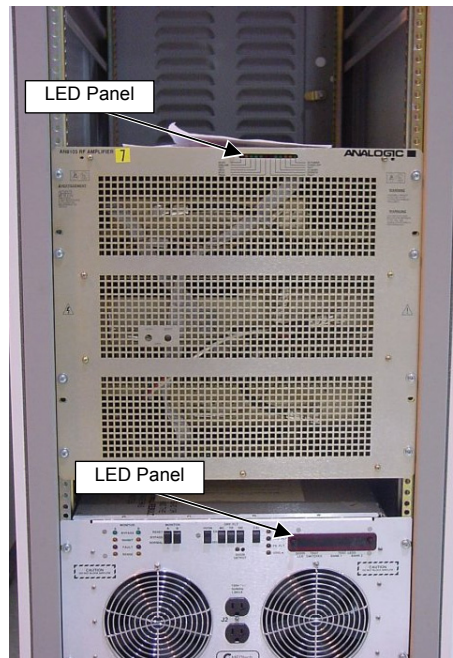
3. Close the plexi-glass door and lock and tag it out. See Illustration 1-2. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)



LOCKOUT AND TAGOUT OF BREAKERS
ILLUSTRATION 1-2

4. Remove the front cover from the SRFD2 Cabinet.

5. Verify all LEDs are OFF (not illuminated) on front of SSM. If the DCHV LED is ON, on the front of the SRFD2 Module, there is still charge stored in the High Voltage power supply capacitors. Wait till all LEDs are OFF. See Illustration 1-3.



SRFD2 LED LOCATION
ILLUSTRATION 1-3

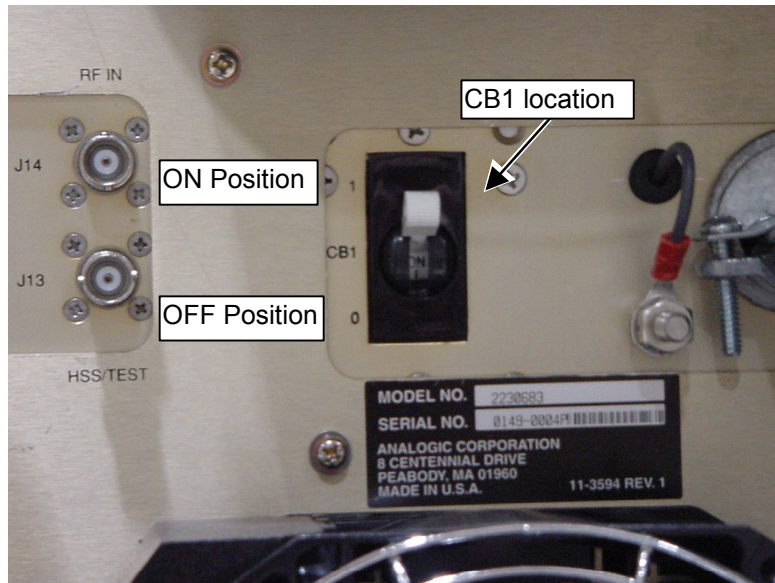
Note

The front panel grate may appear different on newer SRFD2 modules. See Illustration 1-4.



NEWER SRFD2 MODULE
ILLUSTRATION 1-4

6. At the rear of the SRFD2 module, place the CB1 breaker in the OFF position (lever down). See Illustration 1-5 for location of breaker.



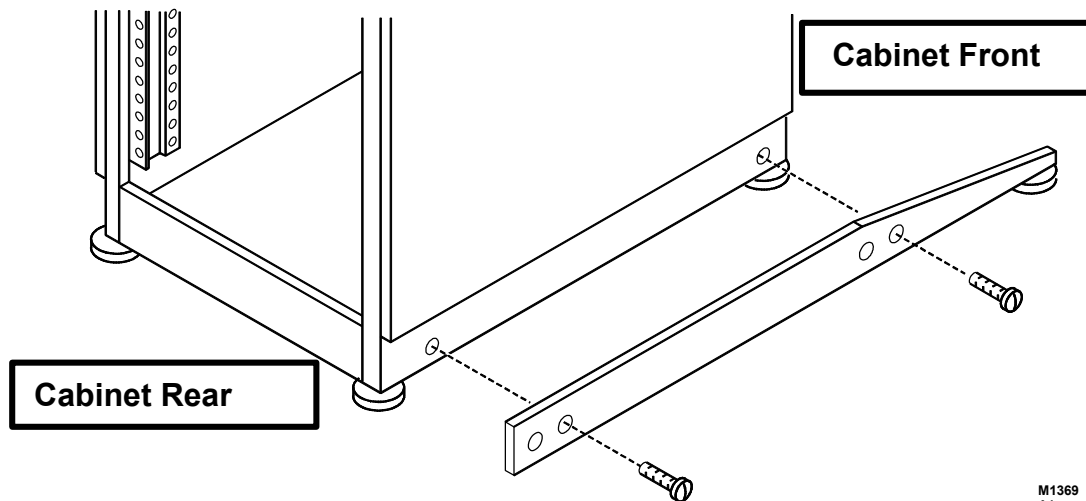
CB1 LOCATION
ILLUSTRATION 1-5

3- INSTALLATION OF RF CABINET ANTI-TIP LEGS

DANGER!!

CABINET TIP HAZARD! POSSIBLE PERSONAL INJURY! CONFIRM THAT THE ALUMINUM ANTI-TIP LEGS STORED IN THE REAR OF THE CABINET ARE INSTALLED ON BOTH SIDES OF THE BASE OF THE CABINET BEFORE ATTEMPTING TO SERVICE THE RF AMPLIFIER.

Using the screws included with the anti-tip legs, secure the aluminum anti-tip legs to the threaded holes on each side of the base of the SRFD2 Cabinet. See Illustration 3-1.



INSTALLING ANTI-TIP LEGS ON SIDES OF CABINET
ILLUSTRATION 3-1

4- SRFD2 FAN REPLACEMENT

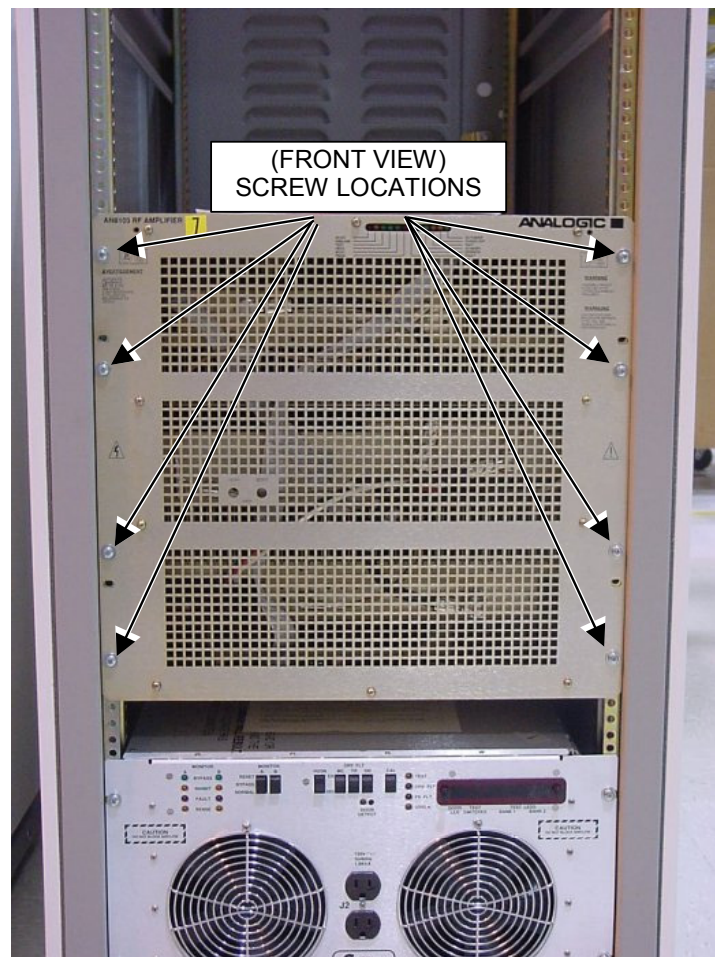
Early RF amplifiers with the serial numbers 0001 to 0030 (See **Section 1 – Overview**) require a fan replacement process that is different than that used for later RF amplifiers. Use **Section 4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY** to replace fans on these amplifiers. Use **Section 4-2 Fan Replacement for Later SRFD2 Amplifiers ONLY** to replace fans on an RF amplifier with a serial number of 0031 or later. See **Section 1 – Overview** for any questions about determining which serial numbers apply. The RF amplifier serial number is marked on a factory sticker attached to the rear of the RF amplifier.

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY



POSSIBLE PERSONAL INJURY! CONFIRM THAT ANTI-TIPS LEGS HAVE BEEN INSTALLED ON EACH SIDE OF CABINET AND THAT LOCKOUT-TAGOUT PROCEDURES HAVE BEEN COMPLETED BEFORE PROCEEDING TO SERVICE THE SRFD2 RF AMPLIFIER.

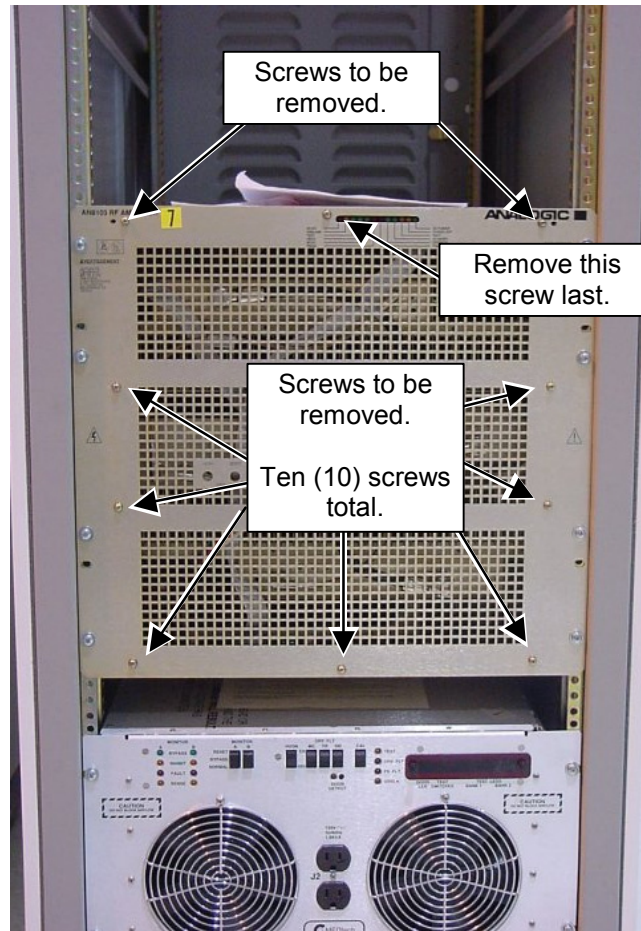
1. Remove the eight screws that secure the amplifier to the cabinet. See Illustration 4-1.



SCREW LOCATIONS ON CABINET
ILLUSTRATION 4-1

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

2. Remove the ten (10) screws to remove the front grill plate of the SRFD II Module. See Illustration 4-2.



LOCATION OF SCREWS TO BE REMOVED
ILLUSTRATION 4-2

Note

Make sure cables connected to the rear of the module are not tangled to protect the cables from damage when pulling out the module.

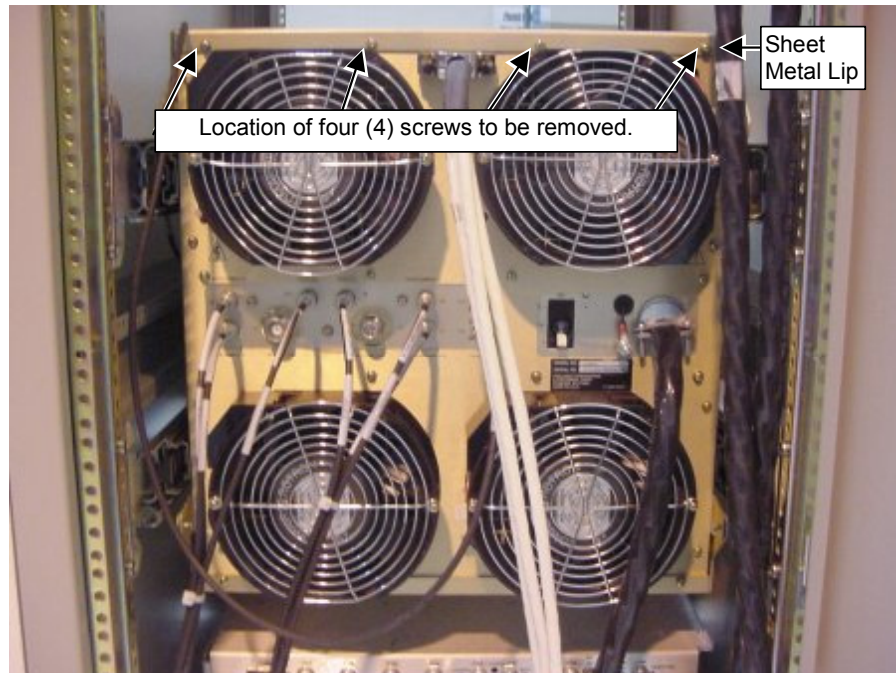
3. Pull module out slightly on the rails to access the rear panel screws.

Note

In order to access all the rear screws, it may be necessary to slide the module in slightly.

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

4. If replacing one of the two upper fans then, from the rear of the SRFD2 module, remove the four (4) screws on the upper sheet metal lip on the module. See Illustration 4-5.



LOCATION OF SCREWS TO BE REMOVED
ILLUSTRATION 4-5

5. If replacing one of the two lower fans then, from the rear of the SRFD2 module, remove the four (4) screws on the lower sheet metal lip on the module. See Illustration 4-6.



LOCATION OF SCREWS TO BE REMOVED
ILLUSTRATION 4-6

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

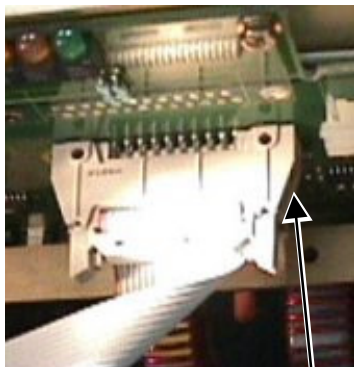
6. After removal of the rear panel screws, slide the module fully forward to access the side screws.
7. If one of the **upper** fans has failed then remove the 5 flathead countersunk screws (Total of 10 Screws) from each side of the upper part of the assembly as shown in Illustration 4-7.



LOCATION OF SCREWS ON SIDE OF MODULE FOR UPPER FAN REPLACEMENT
ILLUSTRATION 4-7

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

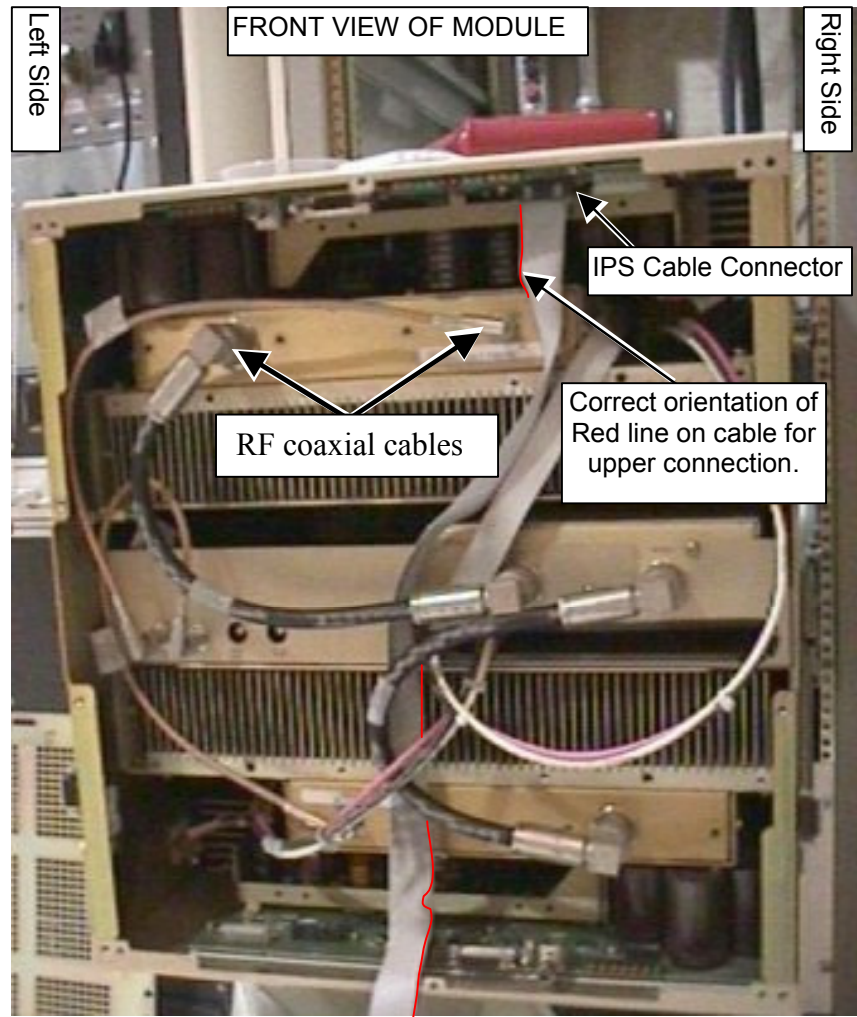
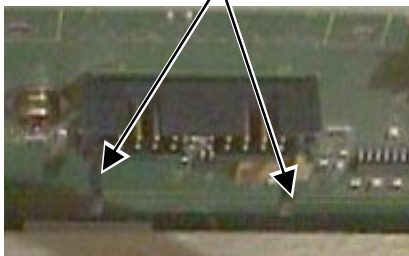
8. Put the Electric-static Discharge Wristband on yourself and attach the ground clip to the cabinet.
9. Disconnect the IPS Cable and RF coaxial cables.
 - a. If replacing a failed upper fan then disconnect the IPS Cable from the upper shelf only. See Illustration 4-8.



Alternate style IPS Cable Connector.

To release, push levers outward.

To release, push slide latches back.

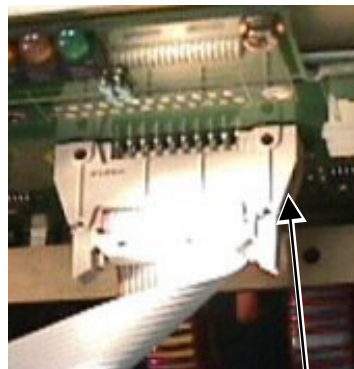


IPS CABLE LOCATIONS
ILLUSTRATION 4-8

- b. If replacing a failed upper fan then disconnect the two RF coaxial cables from the upper shelf and move them out of the way. See Illustration 4-8.

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

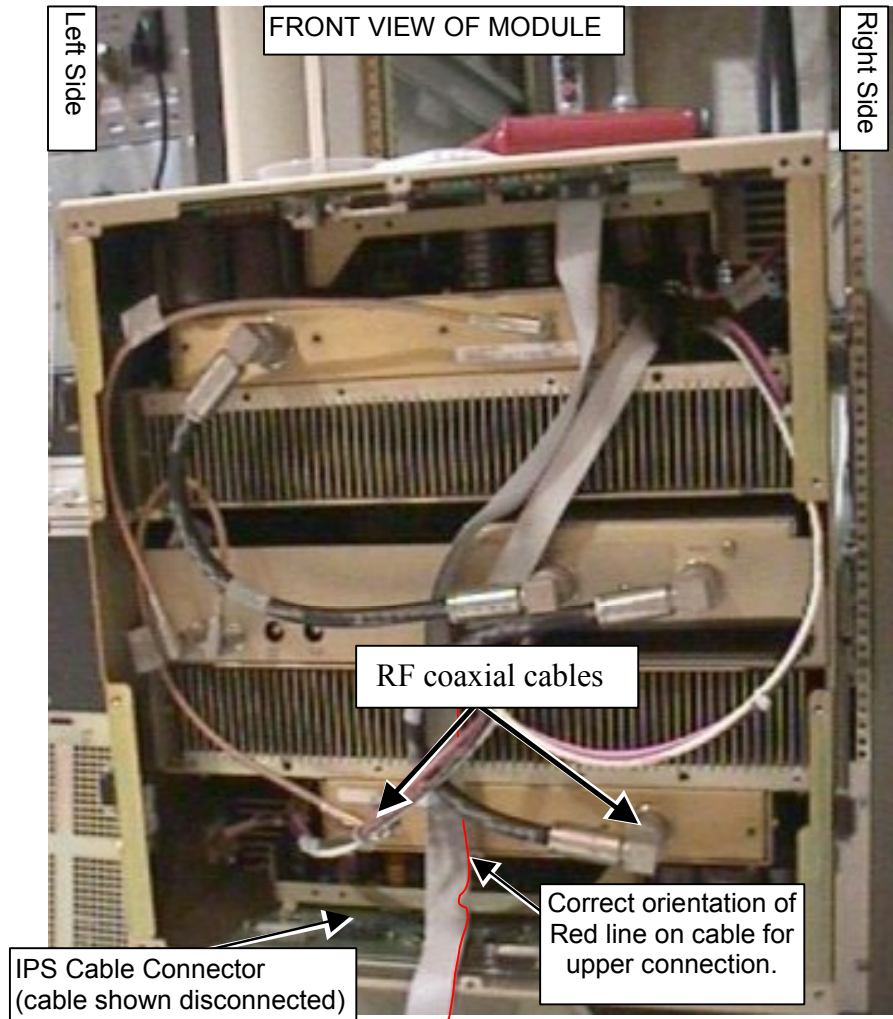
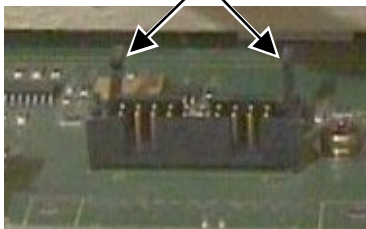
- c. If replacing a failed lower fan then disconnect the IPS Cable from the lower shelf only. See Illustration 4-9.



Alternate style IPS Cable Connector.

To release, push levers outward.

To release, push slide latches back.

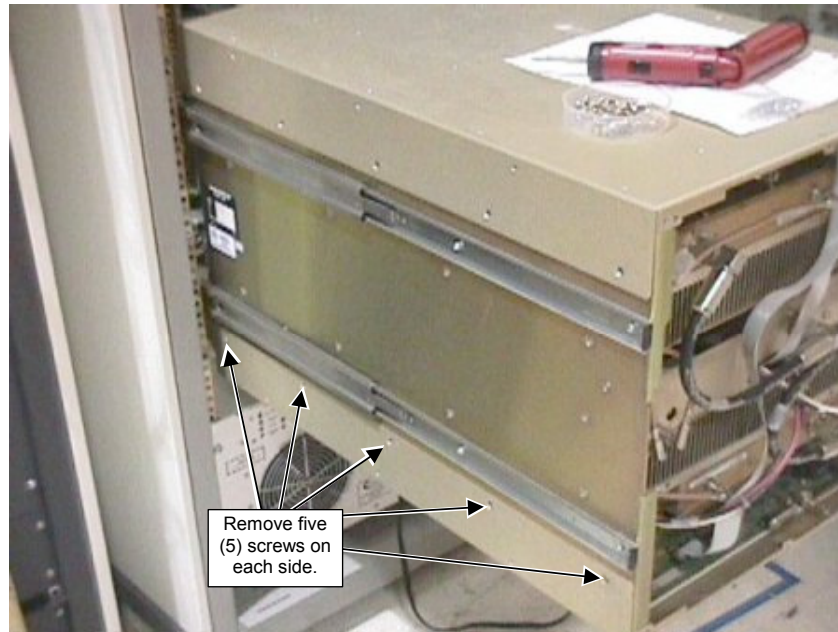


IPS CABLE LOCATIONS
ILLUSTRATION 4-9

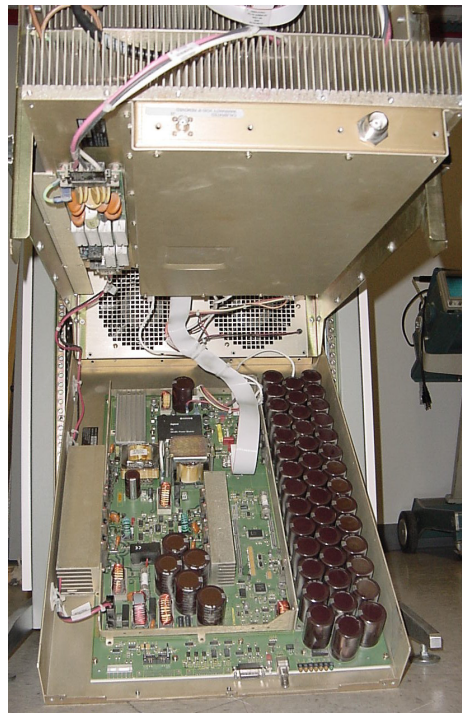
- d. If replacing a failed lower fan then disconnect the two RF coaxial cables from the upper shelf and move them out of the way. See Illustration 4-9.

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

10. If replacing one of the **lower** fans then support the front of the lower half of the module and remove the 5 rear-most flathead countersunk screws (total of 10 screws) from each side of the lower part of the assembly. Allow the front half of the module to hinge down and rest on the floor. See Illustrations 4-10 and 4-11.



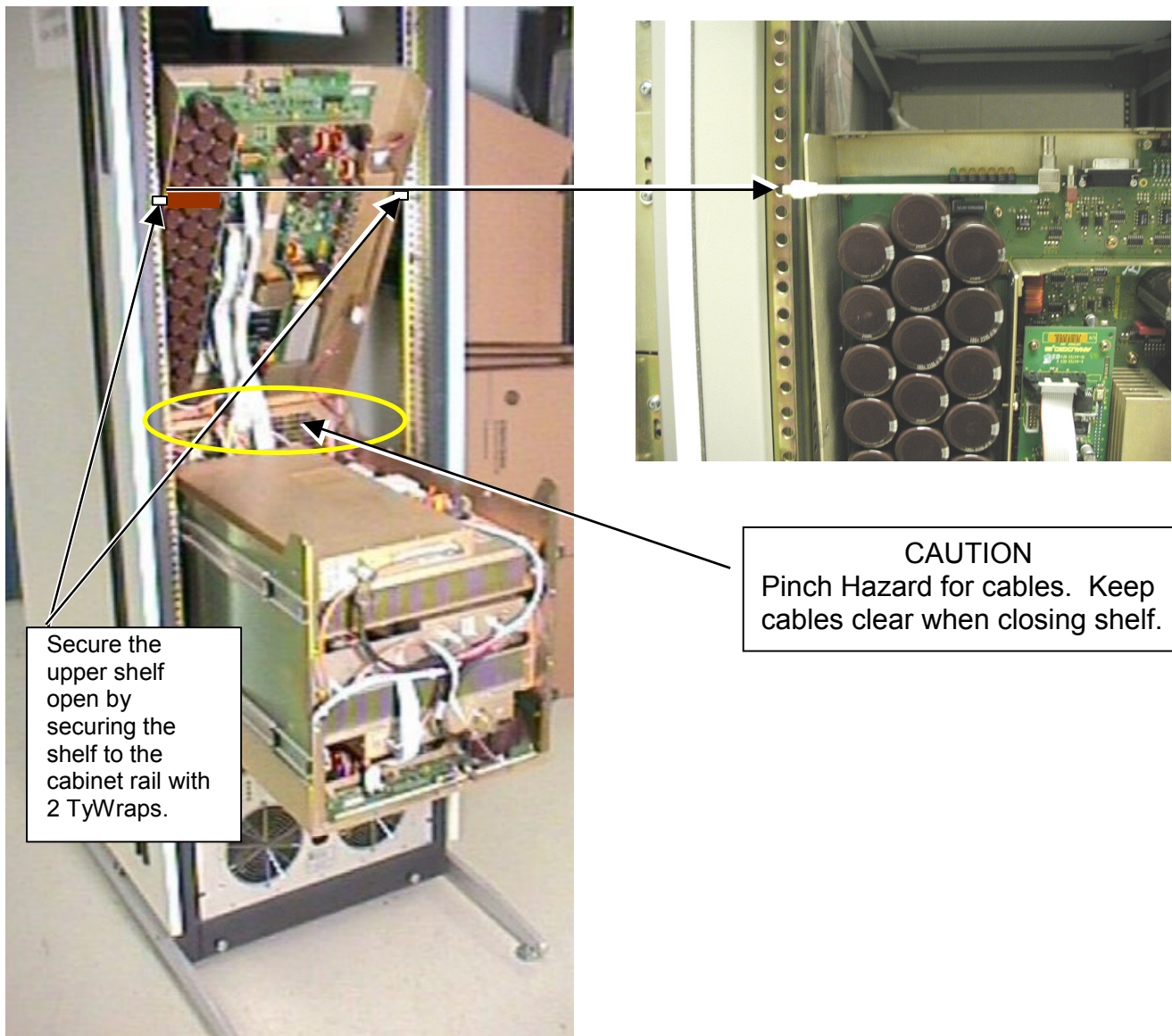
LOCATION OF SCREWS ON SIDE OF MODULE FOR LOWER FAN REPLACEMENT
ILLUSTRATION 4-10



LOWER SRFD2 MODULE EXPOSED
ILLUSTRATION 4-11

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

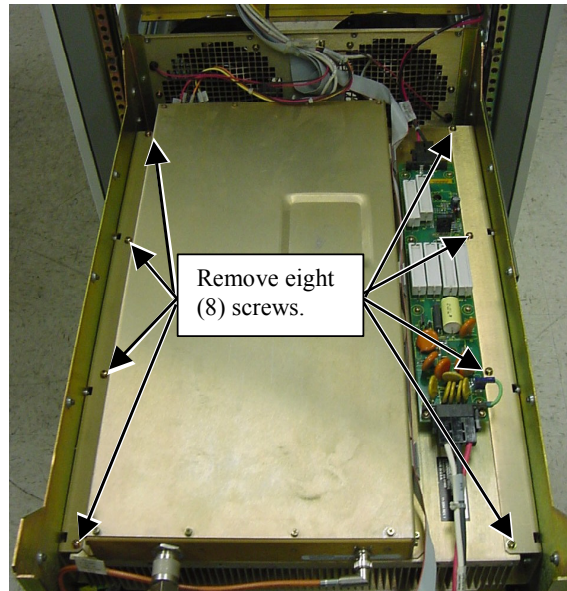
11. If replacing one of the upper fans, open the upper shelf of the module and secure each side to the cabinet frame with two (2) Ty-wraps. See Illustration 4-12.



UPPER SHELF OPEN AND SECURED
ILLUSTRATION 4-12

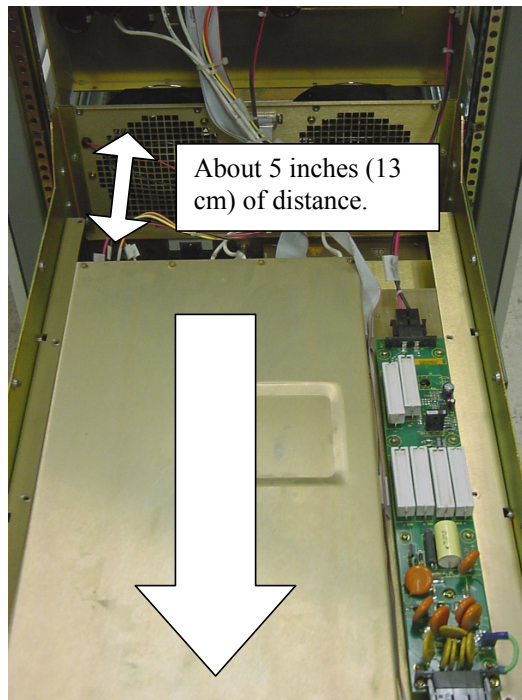
4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

12. If replacing upper fans then remove the 8 screws from the upper module as shown in Illustration 4-13.



8 SCREWS SECURING MODULE TO CHASSIS
ILLUSTRATION 4-13

13. Exercising care to not damage any cables, carefully slide the amp module forward at least 5 inches (~13 cm) to gain access to the fans as shown in Illustration 4-14.



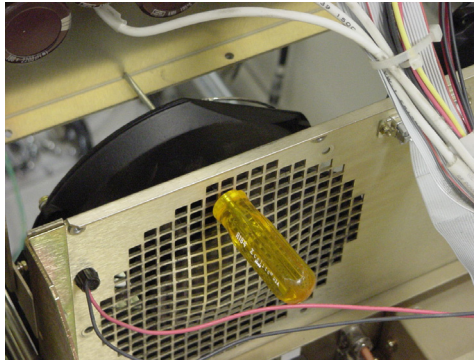
SLIDING AMP MODULE FORWARD TO GAIN ACCESS TO FANS
ILLUSTRATION 4-14

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

Note

Removing the eight (8) screws securing the lower amp module to the chassis when replacing a failed lower fan will not allow the amp module to fall out of the unit.

14. If replacing the lower fans then remove the eight (8) screws in the lower module as shown in Illustration 4-13.
15. Exercising care to not damage any cables, carefully slide the lower amp module forward at least 5 inches (~13 cm) to gain access to the lower fans as shown in Illustration 4-14.
16. Replace the fans.
 - a. If replacing any of the upper fans, insert a slot screwdriver shaft through the fan as shown in Illustration 4-15 to keep the fan from falling into the cabinet after the screws are removed.



USING SCREWDRIVER TO HOLD FAN
ILLUSTRATION 4-15

- b. Remove the screws securing the failed upper or lower fan.
 - c. Use pliers, if necessary, to grasp each red and black power lead connector (**NOT THE WIRES!**) and remove it from the fan.



The fan power leads are polarized and must be correctly connected to the fan. Reversing the leads will cause the fan to rotate in the opposite direction and this will degrade the cooling capability of the RF amplifier.

17. Note that the fan power connections are polarized. On the replacement fan, attach the red (positive) lead to the fan power terminal marked **+** and attach the black (negative) lead to the fan power terminal marked **-**.
18. Attach the replacement fan to the chassis using the 3 screws removed earlier.

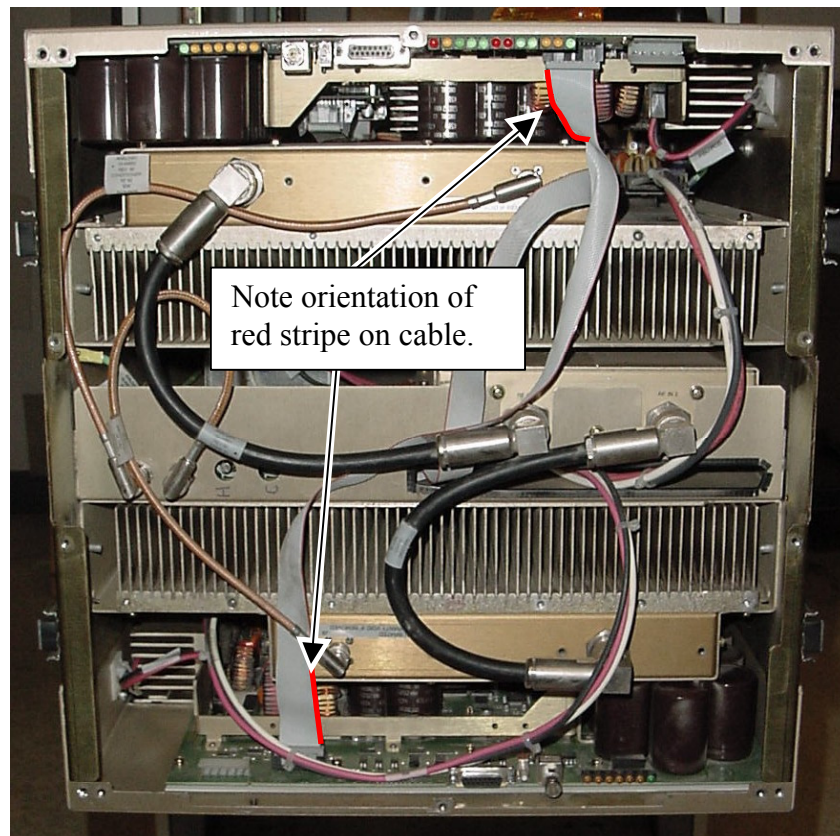
4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

19. Remove the slot screwdriver used to hold the fan in place during the replacement process.
20. Carefully slide either the upper or lower amp module back into the chassis and secure using the 8 screws removed earlier.



The IPS ribbon cable connectors are not keyed. Care must be taken to insure that the connectors are reattached in the correct orientation.

21. Reconnect the IPS ribbon cable so that the red line on the cable is oriented closest to pin 2 (this notation is marked to the side of the connector on the circuit board) on the IPS connector. This is shown in Illustration 4-16.



CABLE PLACEMENT AND ORIENTATION
ILLUSTRATION 4-16

22. Reattach the two (2) RF coaxial cables removed earlier from the front of the module. See Illustration 4-16.

4-1 Fan Replacement for Early SRFD2 Amplifiers ONLY (Continued)

Avoid cable damage! Take care to insure that cables are not pinched and damaged when closing the upper or lower halves of the amplifier assembly.

23. Taking care to insure that cables are not pinched, carefully close the upper or lower half of the amplifier assembly and reinstall the 8 screws to secure it to the chassis.
24. Replace the 4 screws to secure the upper or lower rear sheet metal lip to the chassis.
25. Install the front chassis cover plate on the SRFD2 and secure with the 10 screws removed earlier.
26. Slide the SRFD2 amplifier back into the cabinet and secure to the cabinet using the 8 screws removed earlier.
27. Proceed to **SECTION 5 – RESTORATION**.

4-2 Fan Replacement for Later SRFD2 Amplifiers ONLY



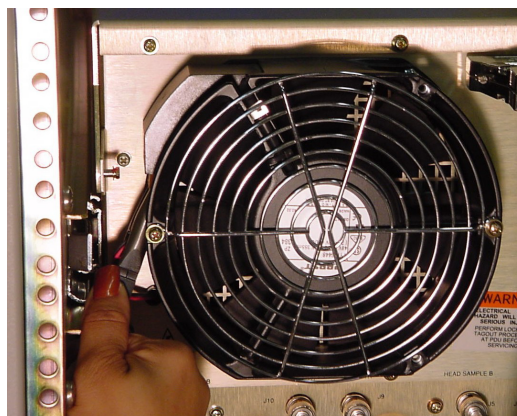
POSSIBLE PERSONAL INJURY! CONFIRM THAT ANTI-TIPS LEGS HAVE BEEN INSTALLED ON EACH SIDE OF CABINET AND THAT LOCKOUT-TAGOUT PROCEDURES HAVE BEEN COMPLETED BEFORE PROCEEDING TO SERVICE THE SRFD2 RF AMPLIFIER.

1. If either of the lower two fan units needs replacement then disconnect all the RF coaxial cables shown in Illustration 4-17. Confirm that the labeling on each cable matches the port to which it is connected before removing the cable.



DISCONNECT RF CABLES
ILLUSTRATION 4-17

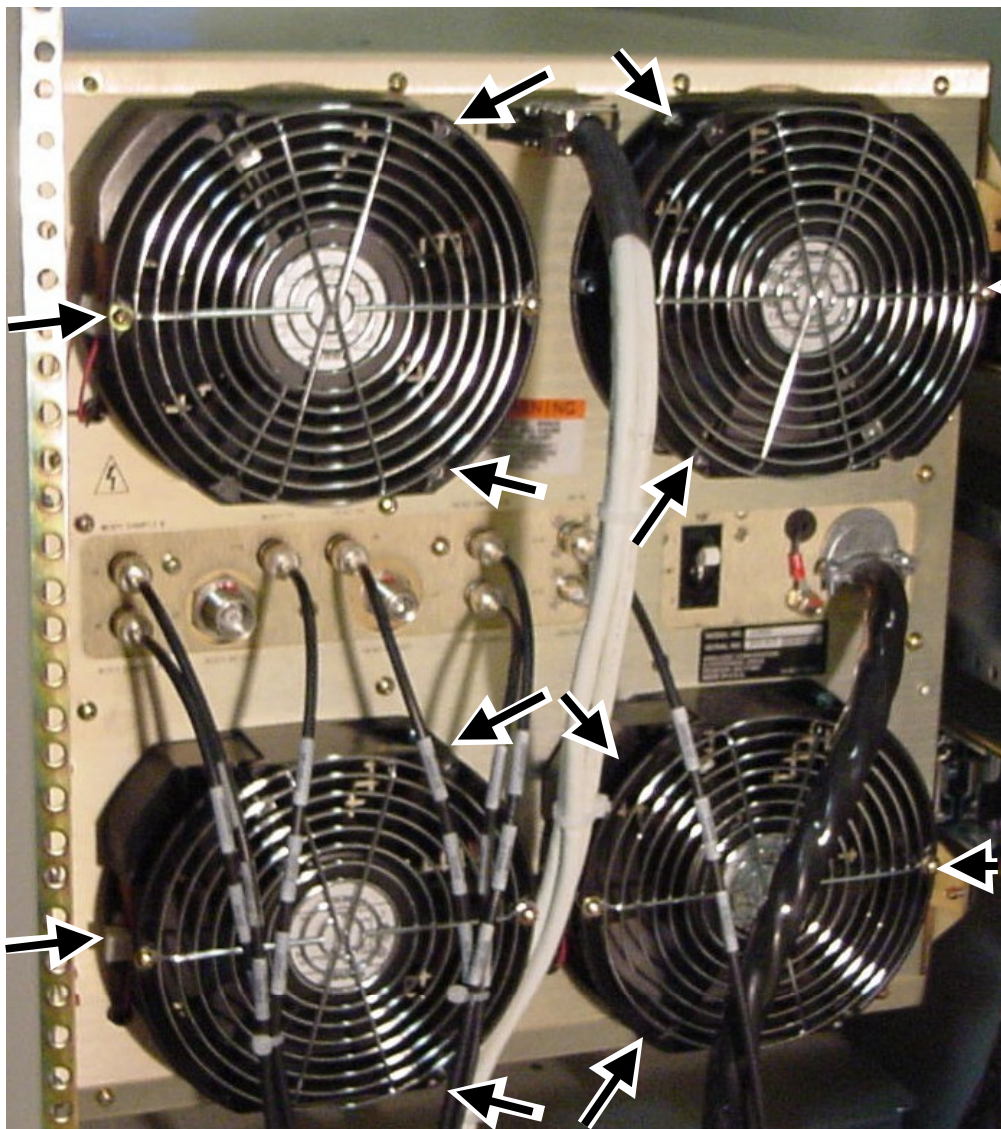
2. Locate the red and black power spade leads to the failed fan. Use needle nose pliers to carefully grasp the lower body of the spade lead connector (**NOT THE WIRE!**) and then pull each spade lead connector loose from the fan. See Illustration 4-17.



FAN POWER LEAD REMOVAL USING NEEDLE NOSE PLIERS
ILLUSTRATION 4-18

4-2 Fan Replacement for Later SRFD2 Amplifiers ONLY (Continued)

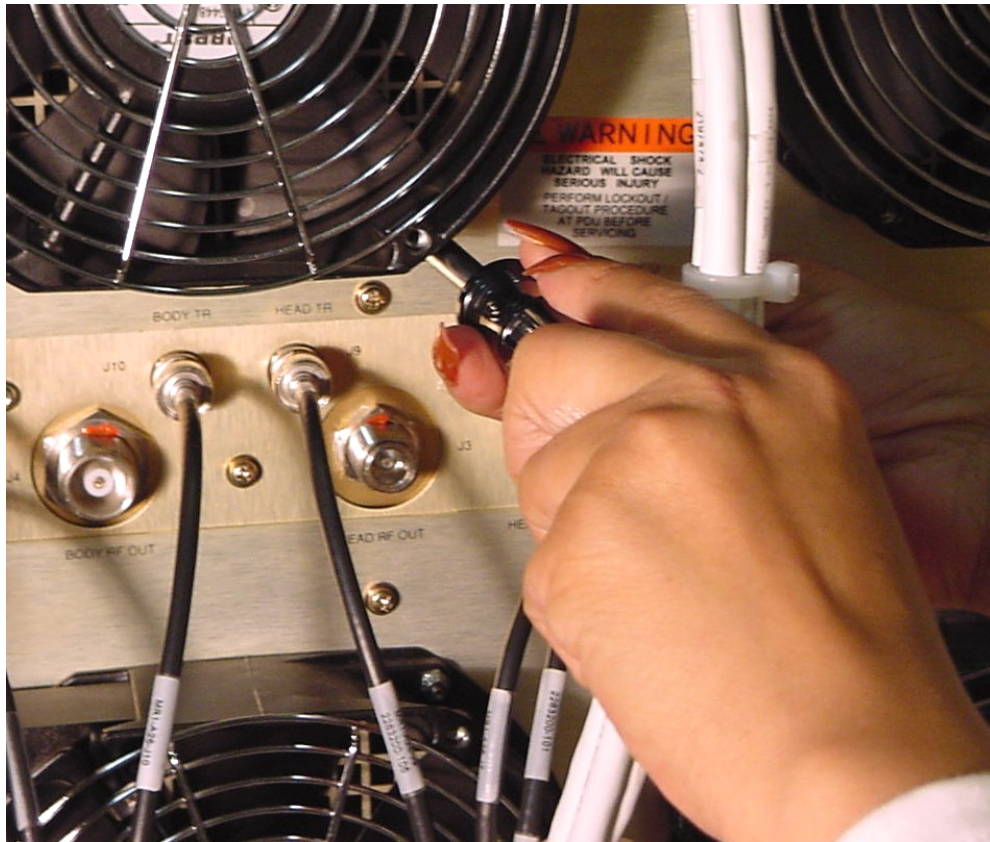
- Each fan is held in place with 3 nuts on 3 threaded studs mounted on the RF amplifier chassis. The location of each of the fan nuts is shown in Illustration 4-19.



LOCATION OF FAN NUTS
ILLUSTRATION 4-19

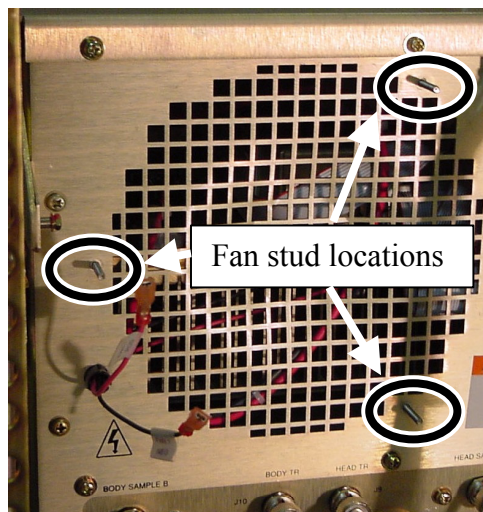
4-2 Fan Replacement for Later SRFD2 Amplifiers ONLY (Continued)

4. Use the 7/32" socket, extension, and drive handle to remove each of the 3 nuts from the failed fan. See Illustration 4-20.



FAN POWER LEAD REMOVAL USING NEEDLE NOSE PLIERS
ILLUSTRATION 4-20

5. Pull the fan away from the RF amplifier. See Illustration 4-21.



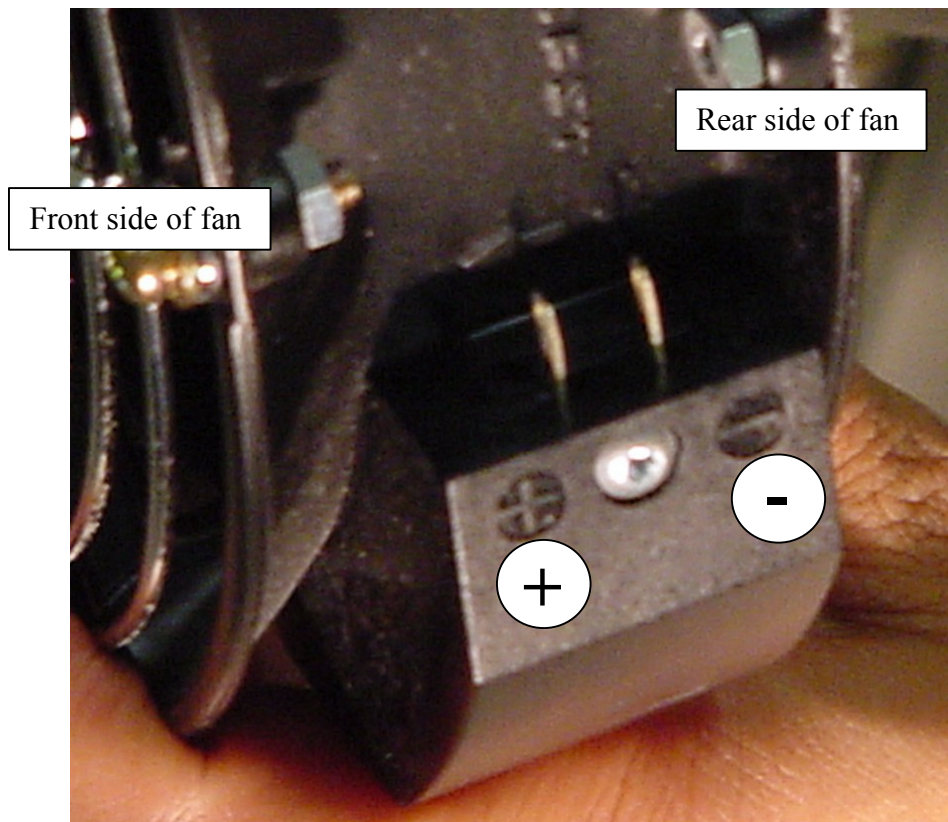
FAN REMOVED FROM RF AMPLIFIER
ILLUSTRATION 4-21

4-2 Fan Replacement for Later SRFD2 Amplifiers ONLY (Continued)

CAUTION

The fan power leads are polarized and must be correctly connected to the fan. Reversing the leads will cause the fan to rotate in the opposite direction. This will degrade the cooling capability of the RF amplifier and may lead to early failure.

6. Obtain the replacement fan and notice that the fan power lead spade connectors are polarized. See Illustration 4-22.



POLARIZED FAN POWER SPADE LEADS

ILLUSTRATION 4-22

7. Before affixing replacement fan to RF chassis, first connect fan power leads to fan so that black (negative) lead connects to rear spade lead on fan while red (positive) lead connects to front space lead on fan.
8. Place replacement fan over threaded studs on RF chassis and **use fingers** to thread nuts one or two turns onto stud. Threading the nuts in this way allows less opportunity for dropping them into the cabinet.
9. Once all nuts are threaded onto studs then use 7/32" socket, extension, and drive handle assembly to tighten nuts and secure fan to RF chassis.

5 – RESTORATION

1. Reconnect any coaxial cables removed earlier to rear of RF amplifier.
2. Remove Lockout/Tagout hardware from PDU and follow proper Lockout/Tagout procedures to put PDU breaker in on position and reapply power to system.
3. Place power breaker on rear of RF amplifier into up (ON) position. Note that fans all rotate in same clockwise direction. If replacement fan does not rotate in correct direction then power leads are swapped. This must be corrected or early amplifier failure may result.
4. Perform one head and body scan to confirm that the system is fully operational.
5. Remove the anti-tip legs from the RF Cabinet.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Oct. 29, 2003	D. Thome	Preliminary Release